

SURAJ YADAV

+91 9305482851 | cs.ersurajyadav@gmail.com | [LinkedIn](#)

PROFESSIONAL EXPERIENCE

An Nextgen Innovation India Private Limited

AI Engineer | Research & Development (Onsite)

25 May 2026 – Present | [Offer Letter](#)

Gurugram, Haryana, India

Core Tech: Python, Ultralytics YOLOv11, OpenCV, Image Annotation, Dataset Engineering, Vision Systems

- Actively spearheading deep learning workflows for the **Stent Detection Machine (SDS Machine)** production deployment at **Meril Life Sciences (MLS-1)**.
- Leading end-to-end dataset development, executing high-precision **micro-defect instance annotations**, and **training custom vision architectures**.
- Managing live system updates and iterations by **continually retraining models on fresh production edge cases** to achieve maximum factory-floor accuracy.
- Developing custom Python and OpenCV infrastructures to process incoming frames and pipe live classification results through **automated checking matrices**.

KP Reliable Technique India Private Limited

Data Analyst (Onsite)

12 Jan 2026 – 20 May 2026 | [Offer Letter](#)

Gurugram, Haryana, India

Core Tech: Data Analytics, Python (Pandas, NumPy, Matplotlib), SQL Databases, Data Visualization, ETL Pipelines

- Engineered and optimized complex **SQL queries and database schemas** to extract, transform, and analyze multi-source operational datasets.
- Programmed custom **Python data analytics scripts** using Pandas and NumPy to automate data extraction, cleaning, and statistical analysis workflows.
- Generated actionable business insights and analytical dashboards to monitor **key performance indicators (KPIs)** and streamline decision-making processes.
- Collaborated with cross-functional teams to integrate structured SQL databases and data pipelines with machine vision and software automation tools.

PROJECTS

Automated Stent Detection System (SDS Machine) – Meril Life Sciences (MLS-1)

Tech Stack: Python, OpenCV, Ultralytics YOLOv11 (Instance Segmentation), PLC Automation, Industrial Lenses, Servo Conveyor System, Strobe Lighting

- Built and trained custom **Ultralytics YOLOv11 instance segmentation models** to automatically detect micro-level stent defects, including **welding defects, pits, heat effects, cracks, and surface irregularities**.
- Engineered an advanced real-time inspection system using a **5 Region of Interest (ROI) configuration** to accurately map, isolate, and categorize **inner, outer, and side defects**.
- Integrated custom Python software with **hardware PLC components**, establishing hardware-software synchronization for **embedded lenses, high-speed lighting triggers, and servo conveyor belts** to enable uninterrupted production-line inspection.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL

Data Analytics & AI: Data Analytics, Pandas, NumPy, Matplotlib, LLM, Agentic AI, OpenCV, Ultralytics YOLOv11 (Segmentation), Machine Vision, Deep Learning

AI Frameworks & Libraries: PyTorch, TensorFlow, NumPy

Web Development: React.js, Next.js, Node.js, Aceternity UI

Tools & Platforms: Git, GitHub, Canva, MS Office

Database & Storage: SQL (PostgreSQL / MySQL), Neo4j (Graph Database for AI)

CS Fundamentals: Artificial Intelligence, DSA, Operating Systems, Computer Networks

EDUCATION

M.G. Institute of Management & Technology

Bachelor of Technology - Computer Science (CGPA: 8.06)

India

2022 - 2026

NIELIT (National Institute of Electronics & Information Technology)

O-Level Certification (Python Programming, IT Tools, Web Technologies & IoT)

India

2026